tracking wildlife movements, supporting ecological conservation and disaster response.

1.2.3 Global influence and market growth

The global impact of drone technology extends beyond technical innovation; it is driving new economic models and reshaping industries. The UAV market, once limited to military procurement, now represents one of the fastest-growing sectors in aerospace and robotics. According to international market forecasts, the global drone market is expected to exceed USD 150 billion by 2035, with significant growth in Asia-Pacific, North America, and Europe. This expansion is fueled by advancements in autonomy, battery life, artificial intelligence, and regulatory modernization that facilitate safe operation within shared airspace.

Governments and private industries are investing heavily in drone research and ecosystem development, including training, manufactur-ing, and maintenance infrastructure. Startups and academic institutions are collaborating on innovations in unmanned traffic management (UTM), beyond-visual-line-of-sight (BVLOS) operations, and AI-driven swarm coordination. These developments are positioning drones as central components of the Industry 5.0 paradigm, where human collaboration, smart automation, and sustainability intersect.

Globally, drones are influencing sectors such as environmental monitoring, renewable energy inspection, border management, and urban logistics, reflecting a shift toward data-driven decision-making and eco-efficient operations. As nations incorporate UAVs into policies and digital strategies, the technology is not merely enhancing operational capacity; it is redefining global connectivity, economic competitiveness, and the future of autonomous mobility.